

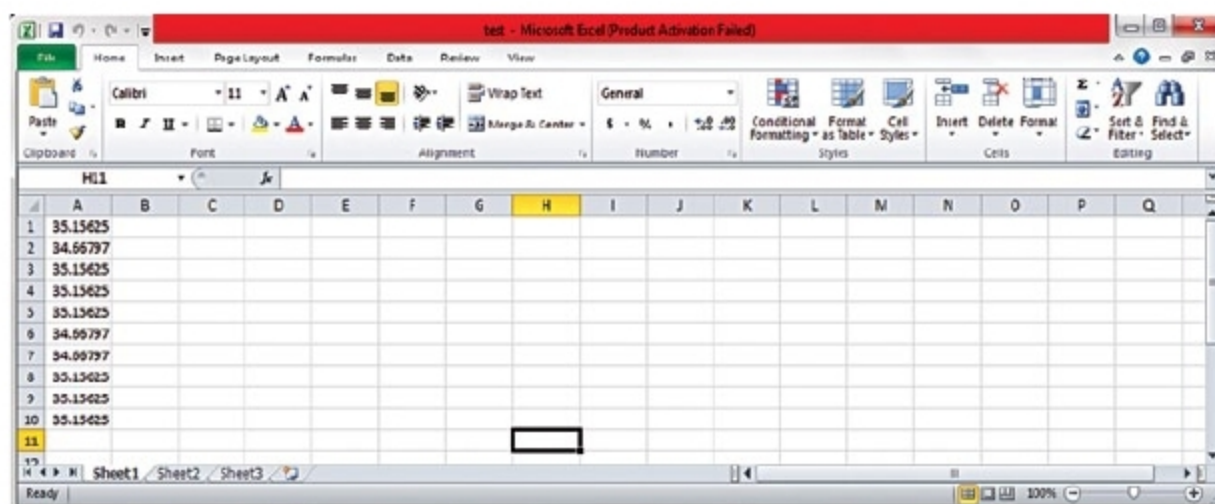


Fig. 3: Screenshot of temperature data saved in Excel

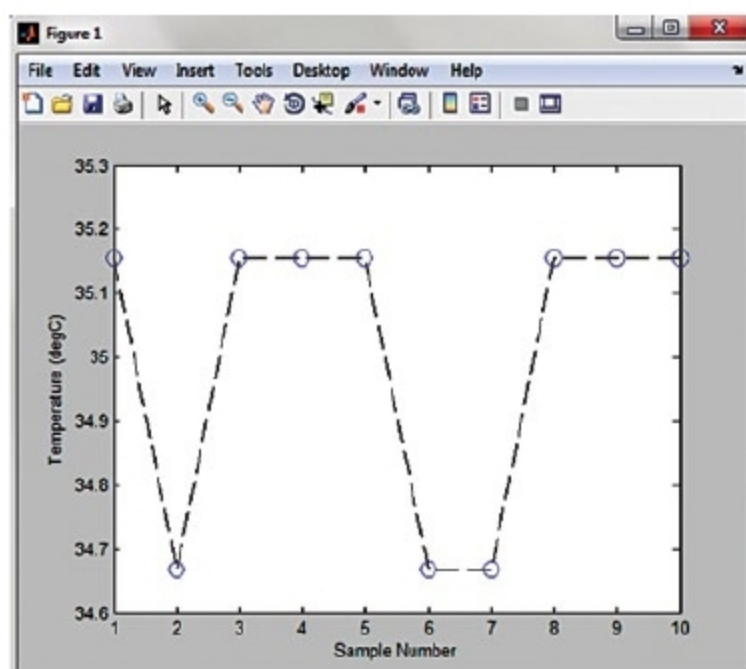


Fig. 4: Screenshot of graph for temperature data

the corresponding port number in the PC where Arduino is installed

Click Run in MATLAB

After pressing Connect in the GUI, MATLAB will try to establish communication with Arduino. While the connection is being made, the GUI displays Connecting.... After attaining successful communication, it displays Connected.

To log data in Excel, press Acquire. Although this program allows acquisition of only 10 samples (for test purposes), you can easily acquire as many samples as you require by changing the line 'for i=1:10' in the source code. For example, if you want to log 100 data samples, modify the assignment statement as 'for i=1:100.' While acquisition occurs, the display shows

Acquiring Sensor Data.

Once data acquisition is complete, the display in the GUI shows Acquisition Complete. Now, you can check the logged data saved in D:\test.xlsx file (you can change this location from the code). A screenshot of the logged data in Excel is shown in Fig. 3.

A graphical representation of the acquired data can be viewed in the figure window of MATLAB by pressing Plot Graph. A screenshot of the graph is shown in Fig. 4.

Further applications

The project can be extended to acquire data from multiple sensors. This simple and effective approach to acquire and log data on an Excel sheet using GUI can help instructors teaching DAQ Systems to practically demonstrate the nuances of the subject matter in class. Since the project is built using a powerful technical computing tool (MATLAB), EFY readers are encouraged to incorporate various data analysis features in the project. **EFY**

Shibendu Mahata is M.Tech (gold medallist) in instrumentation and electronics engineering from Jadavpur University. Currently, he is pursuing PhD from NIT, Durgapur. He has keen interest in MCU-based real-time embedded signal processing and process control systems



Saikat Patra is passionate about electronics and MCU-based embedded system applications

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

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